

Optimization as the Foundation of Machine Learning

Gradient and Hessian

- **Gradient** $\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right)^\top$ points in the direction of steepest ascent and is perpendicular to level sets. Lecture example $f = 3x_1^3 + 2x_1x_2^2 + 4x_2^2$: $\nabla f(1, 1) = (11, 12)$.
- **Hessian** $H = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}\right]$ is symmetric when the second partials are continuous (Schwarz). Lecture: $H_{11} = 18x_1$, $H_{12} = 4x_2$, $H_{22} = 4x_1 + 8$, $\det H(1, 1) = 200$. A quadratic has a constant H .
- **Hessian of a polynomial**: a square cx_i^2 gives $H_{ii} = 2c$; a cross term cx_ix_j gives $H_{ij} = H_{ji} = c$; variables that never meet give 0.
- **Directional derivative** $\frac{\partial f}{\partial s}(x) = \lim_{\alpha \rightarrow 0^+} \frac{f(x+\alpha s) - f(x)}{\alpha} = s^\top \nabla f(x)$, a rate per unit of α . D2: $s = (1, -1)$, $\nabla f = (2, 4)$, $s^\top \nabla f = -2$; per unit length $-\sqrt{2}$; steepest descent $-\|\nabla f\| = -2\sqrt{5}$.

Taylor's formula

- **Taylor**: $f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H(x) h + o(\|h\|^2)$. The linear term changes sign with h , the quadratic term does not.
- $h^\top H h = H_{11}h_1^2 + 2H_{12}h_1h_2 + H_{22}h_2^2$: for the lecture $H(1, 1)$ and $h = (1, 1)$ it is 38.
- **Accuracy**: from $(1, 1)$ along $t(1, 1)$ with $t = 0.1$ the linear model $9 + 23t$ is off by 0.195, the quadratic $9 + 23t + 19t^2$ by 0.005.
- **Quadratic** f : the formula is exact. $q = x_1^2 + x_1x_2 + 2x_2^2$, $x = (1, 2)$, $h = (0.3, -0.2)$ give 10.51 both ways.

Definiteness

- Symmetric A : $A \succ 0$ if $h^\top A h > 0$ for all $h \neq 0$ (a bowl); $A \succeq 0$ if $h^\top A h \geq 0$ (a bowl or a trough); $A \prec 0$, $A \preceq 0$ likewise (a dome); **indefinite** if $h^\top A h$ takes both signs (a saddle).
- **Eigenvalues decide**: all > 0 , all ≥ 0 or mixed signs. For 2×2 : $\lambda^2 - (A_{11} + A_{22})\lambda + \det A = 0$.

A1: $\lambda^2 - 4\lambda + 3 = 0$, $\lambda = (1, 3)$; C2: $\lambda^2 - 4\lambda - 12 = 0$, $\lambda = (-2, 6)$.

- **Sylvester**: $A \succ 0 \iff \Delta_k > 0$ for all k ; $A \prec 0 \iff (-1)^k \Delta_k > 0$ for all k . A3: $\Delta = (2, 4, 22)$, so $H \succ 0$.
- **Semidefinite**: leading minors cannot tell. Use all principal minors (7 of them for 3×3) or eigenvalues with a tolerance.

Minima and feasible directions

- **Local minimum** $x^* \in \Omega$: for some $\varepsilon > 0$, $f(x) \geq f(x^*)$ for all $x \in \Omega$ with $\|x - x^*\| < \varepsilon$. **Global**: for all $x \in \Omega$. **Strict**: $f(x) > f(x^*)$ for $x \neq x^*$.
- **Interior point**: a small ball around it lies in Ω ; otherwise a **boundary** point. Quadrant $x \geq 0$: $(3, 5)$ is interior, $(0, 2)$ is on the boundary.
- **Feasible direction** $s \neq 0$ at x : $x + \alpha s \in \Omega$ for all $\alpha \in [0, \alpha_0]$ with some $\alpha_0 > 0$. D1 at $(0, 2)$: $s_1 \geq 0$; at the corner $(0, 0)$: $s \geq 0$; at an interior point: every s .

Optimality conditions

- **First order, necessary**: $s^\top \nabla f(x^*) \geq 0$ for every feasible s ; at an interior point this is $\nabla f(x^*) = 0$, a **stationary point**. One violating direction rejects a point (D1); accepting needs all of them.
- **Second order, necessary** (interior point): $\nabla f(x^*) = 0$ and $H(x^*) \succeq 0$.
- **Sufficient** (interior point): $\nabla f(x^*) = 0$ and $H(x^*) \succ 0$ give a strict local minimum. There is **no** sufficient condition of first order.
- **Procedure**: solve $\nabla f = 0$; at each candidate $H \succ 0$ means a minimum, $H \prec 0$ a maximum, indefinite H a saddle, singular semidefinite H : analyse f directly. Call a minimum **global** only by convexity or a bound.

When the second-order test is silent

- Same data, opposite answers: $f_1 = x_1^4 + x_2^4$ and $f_2 = x_1^3 - 3x_1x_2^2$ both have $\nabla f(0) = 0$ and $H(0) = 0$; f_1 has a strict global minimum at 0 (B4), f_2 no extremum (B3).

- **Bound from below**: $f_1 \geq 0 = f_1(0)$. **Restrict to a line**: $f_2(x_1, 0) = x_1^3$ changes sign. **Polar form**: $f_2 = r^3 \cos 3\varphi$, 6 sectors of 60° . A picture is a guide, not a proof.
- **Flat directions**: x_1^2 has a whole line of non-strict minima with $H = \text{diag}(2, 0)$; $x_1^3 + x_2^2$ has $H(0) \succeq 0$ and no extremum (H1).

Convexity

- **Convex set**: $\lambda x + (1 - \lambda)y \in \Omega$ for all $x, y \in \Omega$, $\lambda \in [0, 1]$. **Convex function**: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all such x, y, λ . Every norm ball is convex (C1); a ring is not.
- **Test** ($f \in C^2$, open convex Ω): convex $\iff H(x) \succeq 0$ everywhere; $H \succ 0$ everywhere gives strict convexity. C2: $x_1^2 + x_2^2 + a x_1 x_2$ has $\lambda = 2 \pm a$ and is convex $\iff |a| \leq 2$.
- **Local is global**: for a convex f on a convex Ω every local minimum is global, and on $\mathbb{R}^n$ the condition $\nabla f(x^*) = 0$ becomes sufficient.
- **Least squares** $L(w) = \frac{1}{m} \|Xw - y\|^2$: $H_L = \frac{2}{m} X^\top X \succeq 0$, so every solution of $X^\top X w = X^\top y$ is a global minimizer. Line fit: $w^* = (\frac{5}{6}, \frac{3}{2})$.

Top traps

- A cross term cx_ix_j gives $H_{ij} = H_{ji} = c$: not $c/2$ (the entry of A in $x^\top A x$) and not $2c$ (only squares double).
- Leading minors decide only strict definiteness: $\text{diag}(0, -1)$ and $\text{diag}(0, 1)$ both have $\Delta_1 = \Delta_2 = 0$, yet one is $\preceq 0$ and the other $\succeq 0$.
- Gradient first: $H \succ 0$ at a non-stationary point proves nothing. A2: $H(2, -1) \succ 0$, but $\nabla f = (38, -16)$.
- $s^\top \nabla f$ is a rate per unit of α : compare directions only after dividing by $\|s\|$.
- In code a saddle needs `w.min() < -tol` and `w.max() > tol`; the chained `w.min() < -tol < w.max()` calls `diag(0, -1)` a saddle.